L03: Introduction to Keras and Tensorflow

CSci 560 Deep Learning w/ Python (Chollet) Ch. 3

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L03.1 What is TensorFlow

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What is TensorFlow

- TensorFlow is a Python-based open machine learning platform.
- Primary purpose of TensorFlow is to enable researchers to manipulate mathematical expressions over numerical tensors.
- TensorFlow scales fairly well.
- TensorFlow goes beyond scope of NumPy:
  - Can automatically compute the gradient of any differentiable expression.
  - Can run not only on CPUs but also on GPUs and TPUs.
  - Computation defined in TensorFlow can be easily distributed.
  - TensorFlow programs can be exported to other runtimes.

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L03.2 What is ‘Keras‘

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What is Keras

- Keras is a deep learning API for Python, built on top of TensorFlow.
- Through TensorFlow, Keras can run on top of different types of hardware (GPU, TPU or plain CPU).
  - And can be scaled to thousands of machines.
- Keras is a “high-level” API that prioritizes developer time/experience.
- Supports several ranges of “workflows”.

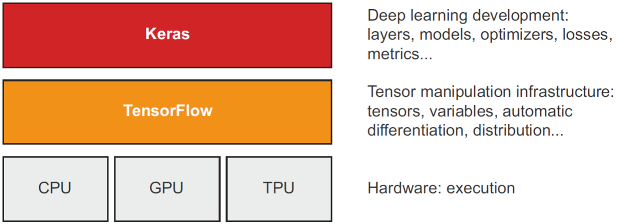

Figure 1: ‘Keras‘ and ‘TensorFlow‘: ‘TensorFlow‘ is a low-level tensor computing platform, and ‘Keras‘ is a high-level deep learning AIP. (Chollet, 2021, pg.70)

L03.4 Setting up a Deep Learning Workspace

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Setting up a Deep Learning Workspace

- To do deep learning on a GPU, you have three options:
  1. Buy and install a physical NVIDIA GPU on your workstation.
     - Dettmer The Best GPUs for Deep Learning in 2023
     - CUDA GPU Compute Capability
  2. Use GPU instances on Google Cloud or AWS EC2.
  3. Use the free GPU runtime from Google Colab: https://colab.research.google.com/

L03.5 First Steps with TensorFlow

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L03.6 Anatomy of a Neural Network: Understanding core Keras APIs

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